

PARCTICAL-02

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved

CODE:-

```
GNU nano 8.7.1                                program-02.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    // Open source file for reading
    source = open("source.txt", O_RDONLY);

    if (source == -1)
    {
        perror("Error opening source file");
        return 1;
    }

    // Open destination file for writing
    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if (destination == -1)
    {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    // Read from source and write to destination
    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead)
        {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }

    if (bytesRead == -1)
    {
        perror("Error reading source file");
        close(source);
    }
}
```

Footer:

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line

```
venu@LAPTOP-KHLR30TF: ~/ + ▼
GNU nano 8.7.1 program-02.c
    perror("Error writing to destination file");
    close(source);
    close(destination);
    return 1;
}
}

if (bytesRead == -1)
{
    perror("Error reading source file");
    close(source);
    close(destination);
    return 1;
}

// Close both files
close(source);
close(destination);

printf("File copied successfully.\n");

return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo     M-6 Copy
```

OUTPUT:-

```
venu@LAPTOP-KHLR30TF:~$ cd os-lab
venu@LAPTOP-KHLR30TF:~/os-lab$ cd practical
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ nano program-02.c
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ echo "Hello, this is my OS lab file." > source.txt
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ls
destination.txt  program  program-01.c  program-02  program-02.c  program.c  source.txt
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ gcc program-02.c -o program-02
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ./program-02
File copied successfully.
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ cat destination.txt
Hello, this is my OS lab file.
venu@LAPTOP-KHLR30TF:~/os-lab/practical$
```